

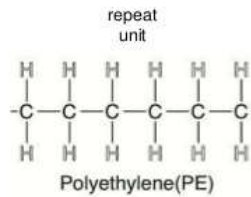
Polymers

Polymer
String Model

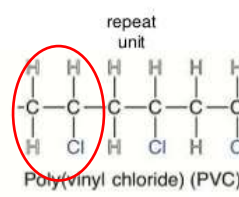
What is a Polymer?

Poly **mer**
Many **Units**

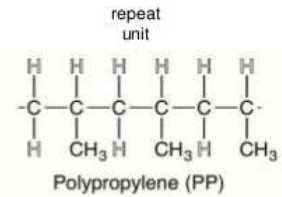
Definition:



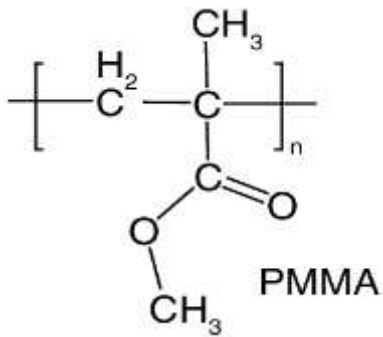
PE Milk Container



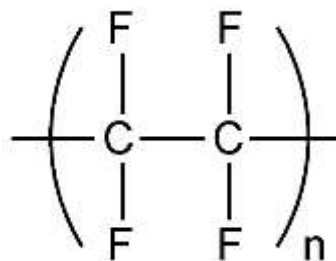
PVC Pipe



PP Rope



Poly(methyl methacrylate)



Poly(tetrafluoroethylene)

Distinguish between Primary and Secondary bonds

Mention the difference in electronegativity causing dipoles, H-bonds.

Molecular Weight

- Number average:

$$\overline{M}_N = \frac{\sum_{i=1}^{\infty} N_i M_i}{\sum_{i=1}^{\infty} N_i}$$

- Weight average:

$$\overline{M}_W = \frac{\sum_{i=1}^{\infty} N_i M_i^2}{\sum_{i=1}^{\infty} N_i M_i}$$

- Polydispersity Index (PDI):

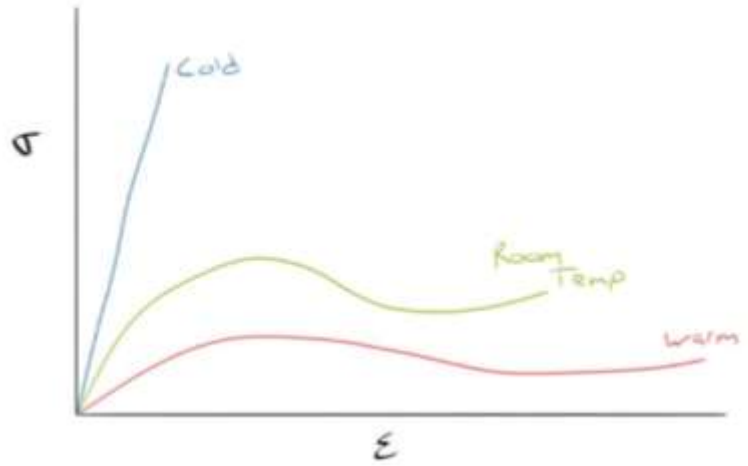
$$PDI = M_w / M_n$$

- N_i = number of molecules with molecular weight M_i

Practice: Calculate M_n and M_w , and the polydispersity index.

Number of Molecules	Mass of Each Molecule
(N_i)	(M_i)
1	800,000
3	750,000
5	700,000
8	650,000
10	600,000
13	550,000
20	500,000
13	450,000
10	400,000
8	350,000
5	300,000
3	250,000
1	200,000

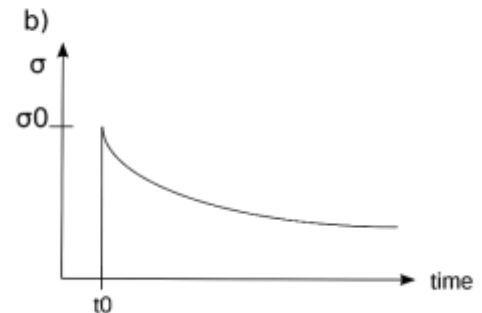
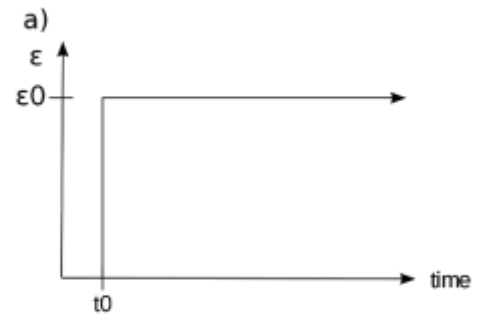
Effect of temperature on tensile test :



When a polymer is subjected to a constant deformation, the stress required to maintain this deformation decreases with time - **Stress Relaxation**

The Modulus is now a function of time
(Relaxation Modulus)

$$E_r(t) = \frac{\sigma(t)}{\epsilon_0}$$



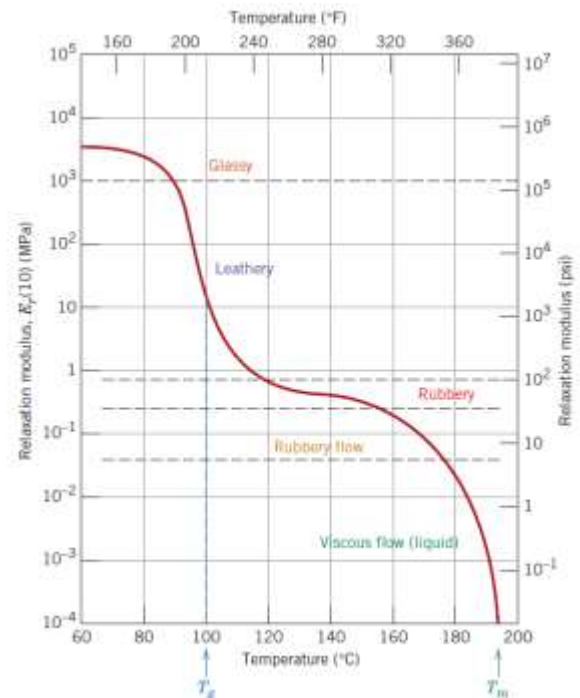
Glass transition vs melting temperature

Glass transition: thermal energy break the secondary bonds in amorphous region.

The temperature at which the amorphous region of a polymer change between the brittle (glassy) condition to a rubbery condition.

(Polymer not necessarily in glass form at room temperature before going through T_g for the first time)

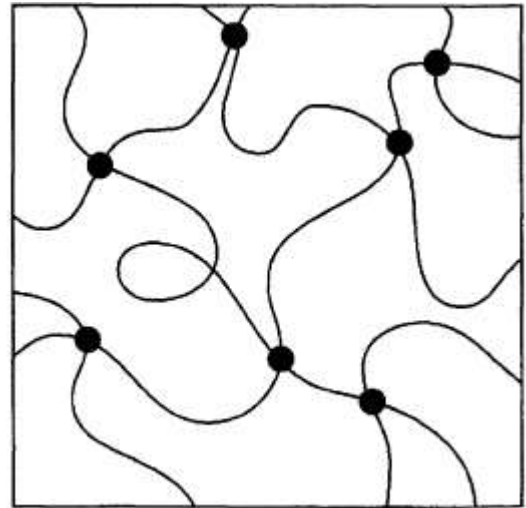
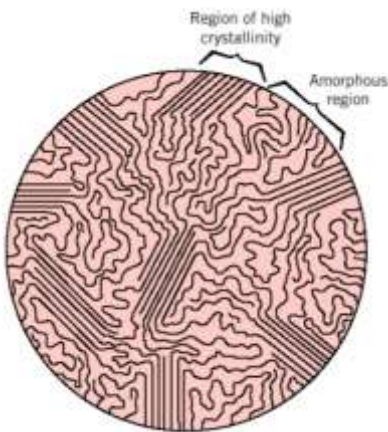
Melting temperature: The temperature at which the secondary bonds break in the crystalline region



Semi-crystalline polymer: Mix of crystalline and amorphous regions

High crystalline polymer: High amount of crystalline regions. Lower effect of T_g on relaxation modulus. Typically higher T_m than semi-crystalline polymers.

Highly crosslinked/network polymer (Figure on the right below): Strong Interaction between the chains, **3D fish net configuration**. Doesn't melt



Learning objectives of the last week (accessible in the syllabus section in Quercus)

Make sure you're familiar with these concepts for your tutorials

Explain the mer unit as the logical building block for understanding polymer structure Recognize or draw and rationalize the relative mechanical behaviour, resistance to chemical dissolution, and softening behaviour upon heating of the following mer units: polyethylene, polypropylene, polyvinylchloride, polytetrafluoroethylene, and polymethylmethacrylate
- Use molecular weight as a means of describing the average length of polymer molecules - Compare the number average molecular weight to the weight average molecular weight and explain the importance of each - Determine both the number average molecular weight and the weight average molecular weight of a polymer sample, provided with the molecular weight distribution data - Recognize the mechanistic similarities between melting, plastic deformation, and chemical dissolution of polymers - Predict the changes to the mechanical properties, softening temperature, and resistance to dissolution resulting from changes to the molecular weight, extent of cross-linking, and the strength of secondary bonds
Differentiate between and infer the property implications of the following microstructural features of polymers: crystallinity, chain length, branching, and extent of cross-linking
Show how differences in electronegativity between atoms in a molecule can create electric dipoles, which in turn can allow secondary bonding
Describe the changes in elastic constant and strength for polymers accompanying changes in temperature close to room temperature
Explain the changes in mechanical behaviour of polymers with small changes in temperature close to room temperature in terms of the relative magnitudes of the thermal energy and the bond energies
Differentiate between the glass transition temperature and the melting temperature of a polymer using an explanation based on microstructure

Questions

1. (1 point) Samples of linear polyethylene (PE), silica window glass (glass), and aluminum (Al) are prepared having identical dimensions. They are each loaded to 50% of their respective yield strength (fracture strength for the glass). The resulting strain is recorded immediately upon loading and then again after 300 days. Which of the following would you expect regarding the strain after 300 days compared to the strain immediately after loading?

- A.** The PE strain will be higher, while the strain for the glass and Al will be unchanged
- B. The strains will be unchanged
- C. The PE and glass strain will be unchanged, while the strain for the Al will be higher
- D. The PE and Al strain will be higher, while the strain for the glass will be unchanged

2. (1 point) Which of the following is the best explanation for the use of the term elastic modulus when referring to polymers, rather than using the term, Young's modulus?

- A.** Polymer elastic behaviour is based on several types of interatomic interaction
- B. Polymers are non-linear elastic
- C. Polymers typically exhibit high strain to fracture
- D. None of the above

3. Which of the following polymers would you expect to have the highest yield strength?

- A. Low molecular weight amorphous polyethylene
- B. Ultra high molecular weight amorphous polyethylene
- C.** Ultra high molecular weight chain oriented polyethylene
- D. Low molecular weight chain oriented polyethylene

6. (1 point) Please determine which of the following options will results in a polymer with a high melting point.

- (a) Increasing molecular weight
- (b) Using branched mer units
- (c) Using a mer unit that will form a network polymer
- (d) Creating cross links between chains

A. a, b, c

B. b, c, d

C. a, c, d

D. b, d

polymers which have branched mer units are irregularly packed so they have low density and so low strength and following that low melting point

Long answers

For some viscoelastic polymers that are subjected to stress relaxation tests, the stress decays with time according to

$$\sigma(t) = \sigma(0) * \exp(-t/\tau)$$

where $\sigma(t)$ and $\sigma(0)$ represent the time-dependent and initial (i.e., time = 0) stresses, respectively, and t and τ denote elapsed time and the relaxation time; τ is a time-independent constant characteristic of the material. A specimen of some viscoelastic polymer with the stress relaxation that obeys the equation displayed above was suddenly pulled in tension to a measured strain of 0.5; the stress necessary to maintain this constant strain was measured as a function of time. Determine $E_r(15)$, the relaxation modulus after 15 s, for this material if the initial stress level was 3.5 MPa, which dropped to 0.5 MPa after 30 s. (4 marks)

$$E_r(15) = \frac{\sigma(15)}{\epsilon_0}$$

$$\sigma(0) = 3.5 \text{ MPa}$$

$$\sigma(30) = 0.5 \text{ MPa} = 3.5 \text{ MPa} * e^{\frac{-30 \text{ s}}{\tau}} \quad +1$$

$$\tau = \frac{-30 \text{ s}}{\ln\left(\frac{0.5}{3.5}\right)} = 15.4 \text{ s} \quad +1$$

$$\sigma(15) = (3.5 \text{ MPa}) \exp\left(-\frac{15 \text{ s}}{15.4 \text{ s}}\right) = 1.32 \text{ MPa} \quad +1$$

$$E_r(15) = \frac{1.32 \text{ MPa}}{0.5} = 2.64 \text{ MPa} \quad +1$$

7. (10 points) Explain why polymers are sensitive to small changes in temperature close to room temperature when metals and ceramics are not. You may use sketches to support your response.

typically the behaviour of polymers are determined by weak, secondary bond called intermolecular bond which can be disrupted very easily by even small changes in temperature. while in metals and ceramics, their behaviour will be determined by strong bond. in metals and ceramics binding energy which holds atoms together is much more than thermal energy.

Multiple Choice Questions

1. (1 point) Which of the following is true of the microstructure of a thermoplastic polymer?
 - A. A slowly cooled thermoplastic may be highly crystalline and a thermoplastic may be completely amorphous**
 - B. A slowly cooled thermoplastic may be completely crystalline
 - C. A slowly cooled thermoplastic may be highly crystalline
 - D. A thermoplastic may be completely amorphous

2. (1 point) Samples of linear polyethylene (PE), silica window glass (glass), and aluminum (Al) are prepared having identical dimensions. They are each loaded to 50% of their respective yield strength (fracture strength for the glass). The resulting strain is recorded immediately upon loading and then again after 300 days. Which of the following would you expect regarding the strain after 300 days compared to the strain immediately after loading?
 - A. The PE strain will be higher, while the strain for the glass and Al will be unchanged**
 - B. The strains will be unchanged
 - C. The PE and glass strain will be unchanged, while the strain for the Al will be higher
 - D. The PE and Al strain will be higher, while the strain for the glass will be unchanged

3. (1 point) Which of the following is the best description for the glass transition temperature?
 - A. The temperature upon heating, when molecules possess enough thermal energy to overcome the secondary, intermolecular interactions within the crystalline regions of the polymer
 - B. The temperature upon heating, when molecules possess enough thermal energy to overcome the secondary, intermolecular interactions within the amorphous regions of the polymer**
 - C. The temperature upon heating, when molecules possess enough thermal energy to overcome the primary, intramolecular bonds within the amorphous regions of the polymer
 - D. The temperature upon heating, when molecules possess enough thermal energy to overcome the primary, intramolecular bonds within the crystalline regions of the polymer

4. (1 point) Which of the following polymers would you expect to have the highest yield strength?
- A. Ultra high molecular weight chain oriented polyethylene**
 - B. Low molecular weight amorphous polyethylene
 - C. Ultra high molecular weight amorphous polyethylene
 - D. Low molecular weight chain oriented polyethylene
5. (1 point) Are the following statements true or false?
- (a) HDPE mostly contains linear polymer chains
 - (b) Increasing chain length decreases the strength of a polymer
 - (c) Poly(methyl methacrylate) (PMMA) can be made 100% crystalline
 - (d) Cross-linked polymers have van der Waals bonds between the chains
- A. True, False, False, False**
 - B. True, True, True, False
 - C. True, False, True, True
 - D. True, True, False, True
6. (1 point) Please determine which of the following options will results in a polymer with a high melting point.
- (a) Increasing molecular weight
 - (b) Using branched mer units
 - (c) Using a mer unit that will form a network polymer
 - (d) Creating cross links between chains
- A. a, b, c
 - B. b, c, d
 - C. a, c, d**
 - D. b, d